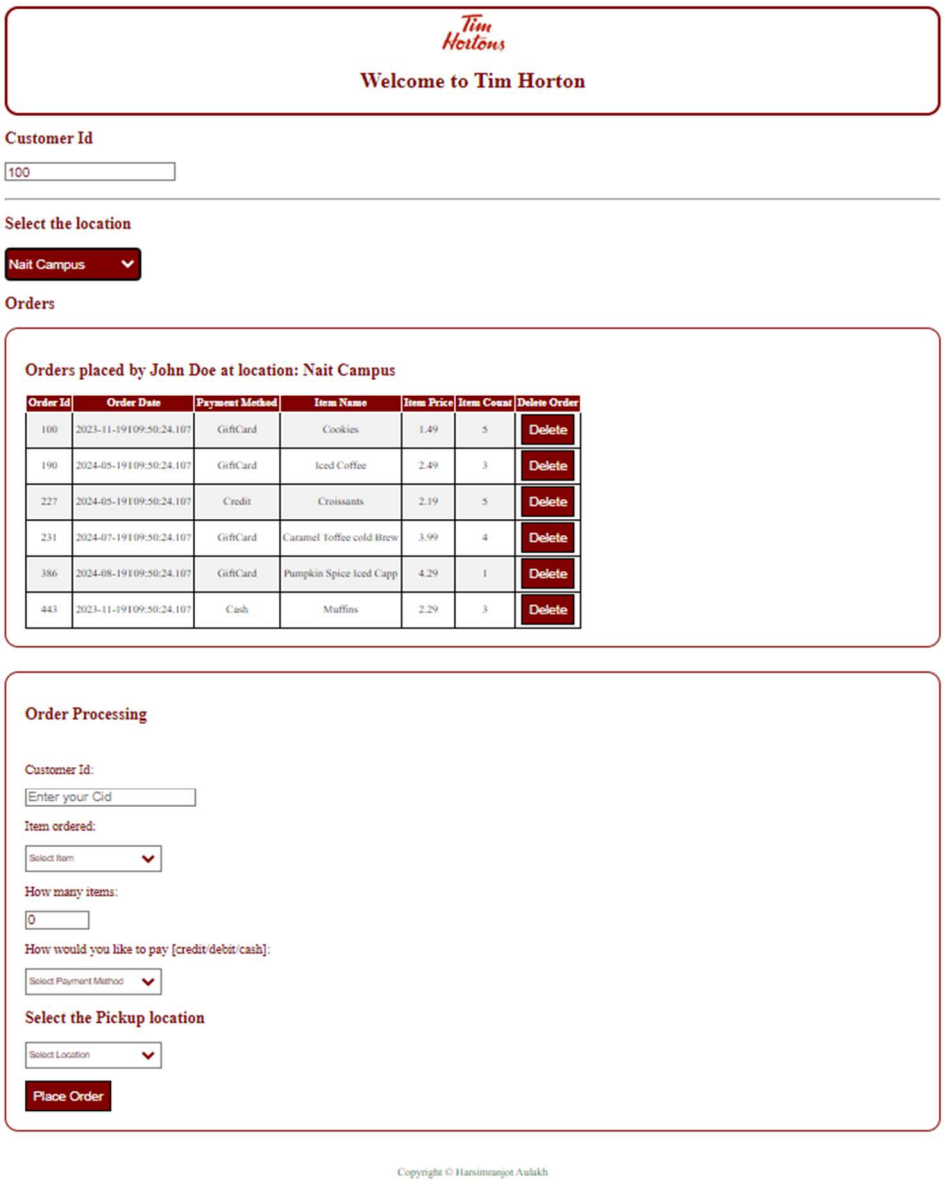
CMPE2550 – ICA11 [ASP] – Data Manipulation

In this exercise, you will incorporate ASP Core functionality, connection to database. You will expand your familiarity with ASP Core and LINQ to execute queries against database, this time completing data manipulation operations.

Tim Hortons requires a basic page from where the user may obtain information regarding orders placed by them at different locations [ASP ICA10], along with order management. The application will allow users to create, update, and delete orders efficiently.

Therefore, to help them you need to design a web solution for this. An example layout and simple style has been shown below, but it is not mandatory to replicate exactly. Feel free to be artistic as you please, but it is advised to focus on the functionality first.



Create a copy of your ASP Assignment10 folder and rename it to ASP Assignment 11 or create a new project. You will require at least 4 items (exact names not required):

index.html This file will be your landing page.

script.js This page will process the actions from the index.html page.

style.css This page will be used to contain style information that may be required by the application logic.

Web Service This will be a service file(s)/ ASP.NET Core project containing data of the application.

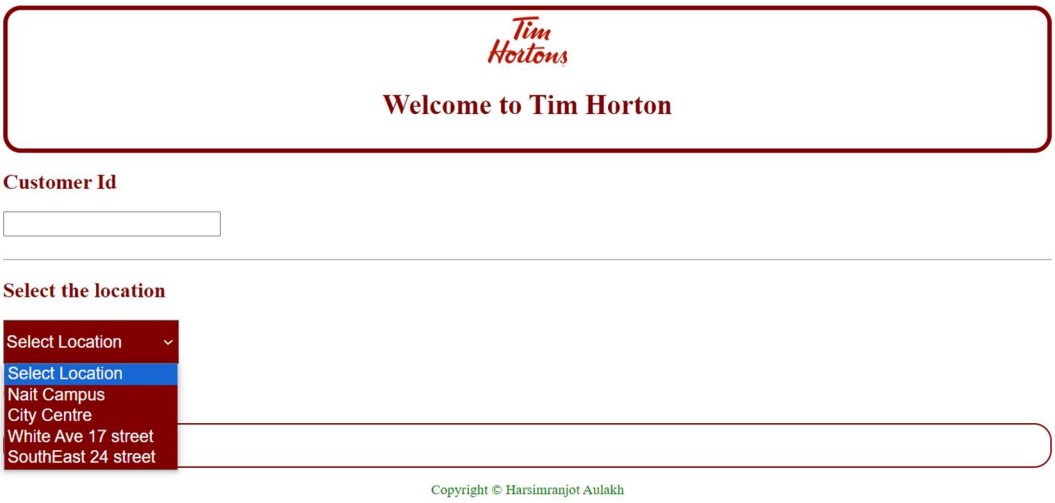
Or you may create an ASP.NET Core project and then create client-side files under a folder.

Restaurant Database

You will have been shown in class how to link your databases into ASP Core files. Use the same copy of the Restaurant Database you created in assignment 10.

Data Retrieval and Display

To retrieve orders, the customer must provide a customer id and select a location. This was completed in assignment 10.



When the user provides a Customer Id **[It is mandatory to provide otherwise give an appropriate alert message and change the focus to Customer Id field]**, and selects a location from the drop-down list, make another call to bring the orders placed by user with provided Customer Id at the selected location from the database. Again, this should have been completed in assignment 10.

Display the order details as shown in the screenshot. Feel free to pick your own design style, but it should not only be data. Also provide a button to user for deleting previous orders. The user is not allowed to update any previous orders.

Orders placed by John Doe at location: Nait Campus

Order Id	Order Date	Payment Method	Item Name	Item Price	Item Count	Delete Order
100	2023-11-19T09:50:24.107	GiftCard	Cookies	1.49	5	Delete
190	2024-05-19T09:50:24.107	GiftCard	Iced Coffee	2.49	3	Delete
227	2024-05-19T09:50:24.107	Credit	Croissants	2.19	5	Delete
231	2024-07-19T09:50:24.107	GiftCard	Caramel Toffee cold Brew	3.99	4	Delete
386	2024-08-19T09:50:24.107	GiftCard	Pumpkin Spice Iced Capp	4.29	1	Delete
443	2023-11-19T09:50:24.107	Cash	Muffins	2.29	3	Delete

You should avoid displaying an empty table when any of the data is missing, or when there are no orders matching a customer ID, and instead just show a status message indicating no records were found, or an appropriate error message.

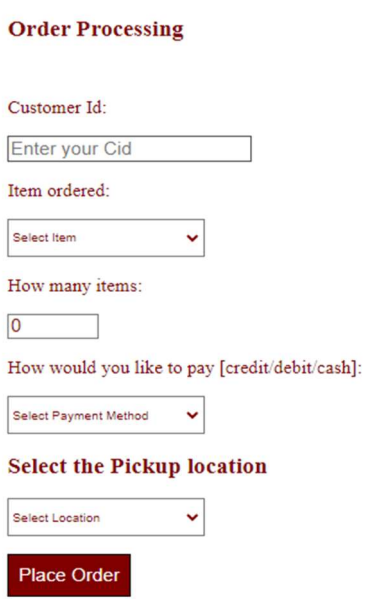
[Delete previous Orders]

The user should be able to delete previously placed orders. When a [Delete] button is clicked, make an Ajax call to delete the order from database.

After deleting the order, update the list on the client side.

[Place new Order]

The next task is to provide a basic form for the customer to make an order that can be processed by the restaurant before the customer picks it up from the pickup location. You must store the order in the appropriate database table. **[It is mandatory to provide valid Customer Id otherwise give an appropriate alert message and change the focus to Customer Id field].** Make sure to check this functionality on both client and Server side.



- Client ID field should be a mandatory field.
- Items ordered and payment method options should be provided in the form of a drop-down list. Fill the item ordered with the data available from server application on page load.
- How many items field should be of number type so that user is not able to enter non-integer values.
- Pickup Location should be a mandatory field.
- Place order button to place an order.

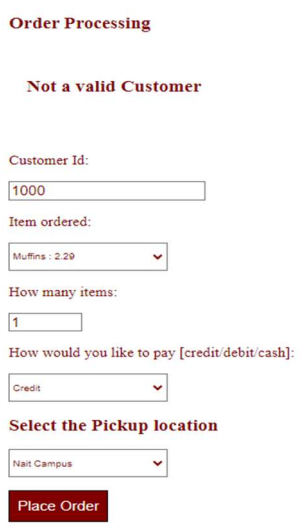
When the user presses the [Place order] Button, the submitted data will be cleaned and examined to ensure everything is provided. An appropriate error message should be displayed if either field is missing. **Perform all the validations both on client and server side.**

If everything is provided, submit the order to the restaurant [server side]. Once the order is processed by the restaurant [means received by the restaurant **[stored in database]** and estimated pick up time is decided], return the processed order details from the restaurant [server] and display it back to the customer for confirmation. Refer to the screenshot provided below. To calculate estimated time for order pickup, generate a random number between 5 and 30 on the server side because the restaurant is preparing your order from 5(less rush) to 30 minutes (peak hours).

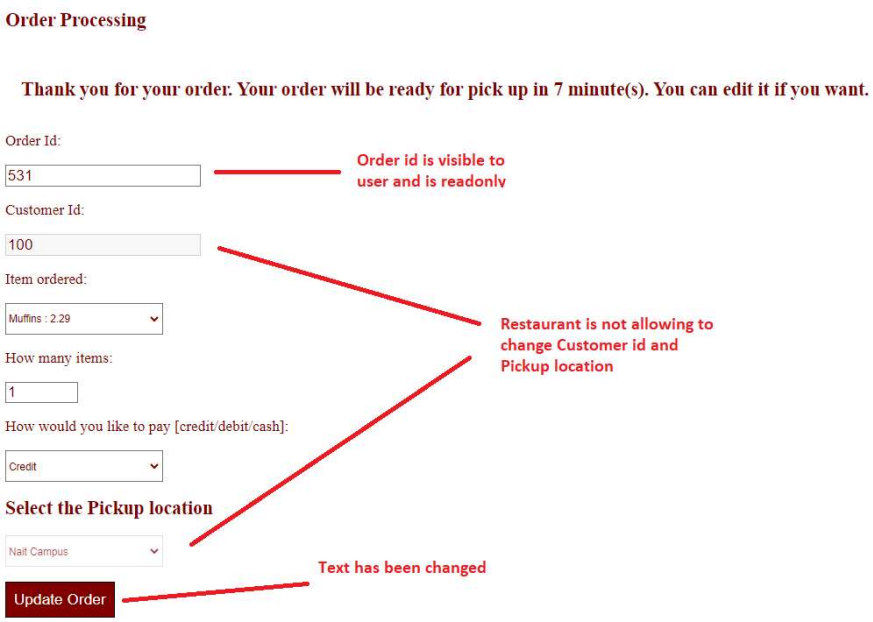
After placing the order, user should be able to update the order details, so change the button to "Update order" from "Place Order".

Possible samples are as follow:

Error Invalid Customer



Successful order placement



[Update Recent Order]

Tim Hortons allows the user to edit order details like number of items, Item, and payment method. The user is not allowed to edit the order id, customer id, or pickup location, **so disable those elements**. If the page is refreshed, then you are not allowed to edit order.

When the user presses the [Update Order] Button, the submitted data will be cleaned and examined to ensure everything is provided. An appropriate error message should be displayed if either field is missing. **Perform all the validations both on client and server side.**

If everything is provided, submit the order to the restaurant [server side]. Once the order is updated in the restaurant database, return a message back to the customer for confirmation. Refer to the screenshot provided below. After this user is not allowed to do any further editing, so disable [Update Order button].

Note: This will be a living document as it is the first time it is being delivered. If typos are encountered, or an explanation is not clear or seems incomplete, please approach your instructor about it so that the feedback may be passed to the document creator. If the fix is complex enough, this document will be added to or changed, and the class alerted. Do keep in mind that you are expected to do some of the thinking and specific design, so this document has not been designed as a step-by-step recipe for completing the assignment, but more intended as a discussion of the requirements.